

ISE 5405: Optimization I

Lecture 4: The Geometry of Linear Programming I

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Polyhedra, Convexity & Corner Points • Bertsimas–Tsitsiklis
[§§2.1–2.2](#)

Where We're Headed (Ch. 2)

- **Today:** polyhedra as intersections of halfspaces; convexity; and three equivalent definitions of a “corner” §§2.1–2.2
- **Next lecture:** standard form, degeneracy, when corners exist, and why an optimum lives at one §§2.3–2.8

The transition

Chapter 1 *showed* us optima at corners. Chapter 2 **proves** it—and hands the simplex method its playground.

Proof habits return

Definitions unpacked carefully, contradiction arguments, and “for all / there exists” read precisely.

The General Optimization Problem

$$\min f(x) \quad \text{s.t.} \quad x \in S,$$

where $f : \mathbb{R}^n \rightarrow \mathbb{R}$ and $S \subseteq \mathbb{R}^n$.

Trichotomy of outcomes

Infeasible means $S = \emptyset$; unbounded means $\forall K \exists x \in S : f(x) < K$; otherwise an optimum x^* satisfies $f(x^*) \leq f(x)$ for every $x \in S$.

General models can evade the trichotomy

For $\min_{x \in \mathbb{R}} e^x$, the infimum is 0 but is never attained: the model is feasible, not unbounded below, and has no optimum.

LP promise

For LPs the trichotomy is exact: infeasible, unbounded below, or an attained optimum.

Local vs Global Optima

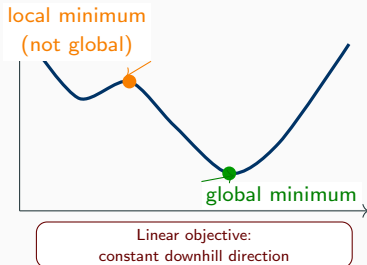
Two notions

Global: $f(x^*) \leq f(x)$ for every $x \in S$.

Local: x^* beats only feasible neighbors within some distance $\epsilon > 0$.

The three properties we will earn

1. An LP is infeasible, unbounded, or has an optimum.
2. Every local LP optimum is global.
3. A nonempty polytope has an optimum at a corner.



Convex Sets

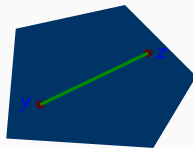
Definition

S is **convex** if, for all $y, z \in S$ and $\lambda \in [0, 1]$,

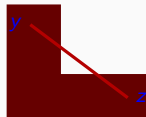
$$\lambda y + (1 - \lambda)z \in S.$$

The whole segment between any two points stays inside.

- $\lambda y + (1 - \lambda)z$ slides from z at $\lambda = 0$ to y at $\lambda = 1$.
- Intersections of convex sets are convex.
- Halfspaces are convex; therefore every polyhedron is convex.



Pentagon: the segment stays inside—consistent with convexity.

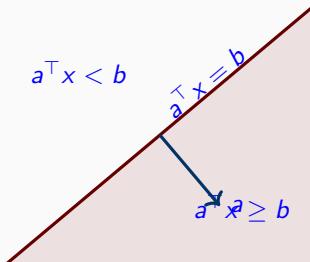


Star and L-shape: a displayed segment leaves the set. They are not convex; one failed pair disproves it.

Hyperplanes & Halfspaces (Def. 2.3)

For $a \in \mathbb{R}^n \setminus \{0\}$, $b \in \mathbb{R}$

- $\{x : a^\top x = b\}$: a **hyperplane**—a line in $\mathbb{R}^2$, a plane in $\mathbb{R}^3$;
- $\{x : a^\top x \geq b\}$: a **halfspace**.
- The hyperplane is the boundary of its halfspace.
- A hyperplane is the intersection of two opposing halfspaces.
- a is perpendicular to the hyperplane.



Polyhedra (Def. 2.1): Intersections of Halfspaces

Definition

A **polyhedron** is a set

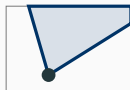
$$P = \{x \in \mathbb{R}^n : Ax \geq b\},$$

the intersection of finitely many halfspaces.

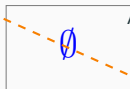
- The feasible set of every LP is a polyhedron.
- Standard form $\{x : Ax = b, x \geq 0\}$ is also a polyhedron: equalities are pairs of inequalities.
- As an intersection of convex sets, a polyhedron is convex.

Definition 2.2

A bounded polyhedron—one that fits in a sufficiently large box—is a **polytope**.



After two cuts:
1 reported interior corner



After cut 3: empty;
later cuts stay empty

Intersection trace

Two cuts leave a nonempty polygon; the third makes the intersection empty.

Three Definitions of "Corner Point"

1. Extreme point

Not a convex combination of other points of P . geometric

2. Vertex

Unique winner of some LP over P . geometric / optimization

3. Basic feasible solution

n linearly independent active constraints. algebraic—computable

Punchline (Thm 2.3)

All three are equivalent. We think geometrically and compute algebraically—the best of both worlds.

Why three? Each definition makes a different theorem easy.
Mathematicians hoard good definitions like tools.

Extreme Points (Def. 2.6)

Definition

$x \in P$ is an **extreme point** if we cannot write

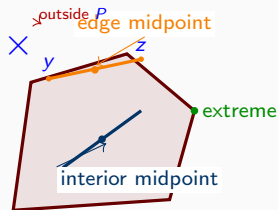
$$x = \lambda y + (1 - \lambda)z$$

with $y, z \in P$, both different from x , and $\lambda \in [0, 1]$.

- This is purely geometric; it does not depend on how P is written.
- Interior points and edge-interior points fail: they are midpoints of neighbors.

Negation practice

To show x is not extreme, exhibit y, z, λ . To show it is extreme, assume such witnesses exist and derive a contradiction.



Reading the picture

The edge and interior points are midpoints of displayed feasible pairs, so they are not extreme; the highlighted vertex is extreme.

Vertices (Def. 2.7)

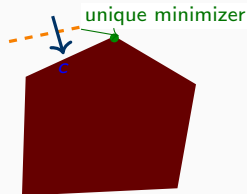
Definition

$x \in P$ is a **vertex** if some cost vector c satisfies

$$c^\top x < c^\top y \quad \text{for every } y \in P, y \neq x.$$

Thus x is the unique optimum of some LP over P .

- A supporting hyperplane touches P only at x .
- In Lecture 2, dragging c made some corner win uniquely.



A level line touches P at exactly one point.

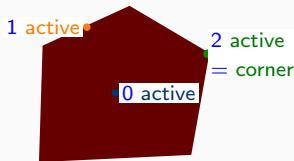
Active Constraints (Def. 2.8)

Definition

Constraint i is **active** (tight, binding) at x^* when it holds with equality:

$$a_i^\top x^* = b_i.$$

- Active constraints are the ones x^* is pressed against.
- In $\mathbb{R}^2$: an interior point has zero active boundary constraints; an edge point has one; a corner has two.
- This bridges geometry and algebra: corners occur where enough constraints are active.



In $\mathbb{R}^n$, n independent active constraints pin a point.

Basic (Feasible) Solutions (Def. 2.9)

Definition

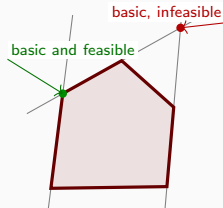
x^* is a **basic solution** if:

1. all equality constraints are active, and
2. among all constraints active at x^* , some n are linearly independent.

If it also satisfies every constraint, it is a **basic feasible solution (BFS)**.

Algebraic meaning

“Linearly independent constraints” means the vectors a_i are linearly independent. n independent active constraints pin down one point.



Two independent lines meet at one point
when $n = 2$.

The Equivalence Theorem (Thm 2.3)

Theorem

For a nonempty polyhedron P and $x^* \in P$, the following are equivalent:

$$x^* \text{ is a vertex} \iff x^* \text{ is an extreme point} \iff x^* \text{ is a BFS.}$$

Why it matters

- BFS seems representation-dependent because it mentions constraints; the theorem says geometry decides.
- We may choose whichever definition makes a proof easiest.

Proof architecture

Prove a cycle:

$$\text{vertex} \Rightarrow \text{extreme} \Rightarrow \text{BFS} \Rightarrow \text{vertex.}$$

Three implications give all six.

Vertex $\Rightarrow$ Extreme Point

x^* is a vertex, so some c satisfies $c^\top x^* < c^\top y$ for every $y \in P \setminus \{x^*\}$. To show that x^* is extreme, argue by contradiction. Suppose

$$x^* = \lambda y + (1 - \lambda)z, \quad y, z \in P \setminus \{x^*\}, \quad \lambda \in [0, 1].$$

1. The endpoints are impossible: $\lambda = 0$ would give $x^* = z$, and $\lambda = 1$ would give $x^* = y$. Thus $\lambda \in (0, 1)$.
2. The vertex property applies to both witnesses: $c^\top y > c^\top x^*$ and $c^\top z > c^\top x^*$.
3. Apply $c^\top$ to the convex combination:

$$c^\top x^* = \lambda c^\top y + (1 - \lambda)c^\top z > \lambda c^\top x^* + (1 - \lambda)c^\top x^* = c^\top x^*.$$

Contradiction

A number cannot be strictly greater than itself. No such y, z, λ exist, so x^* is an extreme point. $\square$

Extreme Point $\Rightarrow$ Basic Feasible Solution

Use the contrapositive. Let $x^* \in P = \{x : Ax \geq b\}$ be feasible but not basic. With active set $I = \{i : a_i^\top x^* = b_i\}$, we have $\text{rank}\{a_i : i \in I\} < n$. We will exhibit two witnesses that show x^* is not extreme.

1. Rank deficiency gives $d \neq 0$ with $a_i^\top d = 0$ for every $i \in I$.
2. For $\varepsilon > 0$, define $y = x^* + \varepsilon d$ and $z = x^* - \varepsilon d$. Active constraints remain equalities at both points.
3. Every inactive constraint has strict slack: $a_i^\top x^* - b_i > 0$. Choose one step size that works for all:

$$0 < \varepsilon < \min_{\substack{i \notin I \\ a_i^\top d \neq 0}} \frac{a_i^\top x^* - b_i}{|a_i^\top d|}.$$

If the index set is empty, any positive ε works. Then all inactive constraints still hold, so $y, z \in P$.

4. Since $d \neq 0$, the points are distinct from x^* , and

$$x^* = \frac{1}{2}y + \frac{1}{2}z.$$

Conclusion. x^* is not extreme, so the contrapositive proves extreme point $\Rightarrow$ BFS. $\square$

Basic Feasible Solution $\Rightarrow$ Vertex

1. Select independent active constraints

Let $I(x^*) = \{i : a_i^\top x^* = b_i\}$. Since x^* is a BFS, choose $B \subseteq I(x^*)$, $|B| = n$, such that $\{a_i : i \in B\}$ is linearly independent. The square matrix A_B is therefore nonsingular.

2. Build an objective from them

Set $c = \sum_{i \in B} a_i = A_B^\top \mathbf{1}$. For every $y \in P$,

$$c^\top (y - x^*) = \sum_{i \in B} (a_i^\top y - b_i) \geq 0.$$

The summands are nonnegative because y is feasible and each selected constraint is active at x^* . Thus x^* minimizes $c^\top x$.

3. Prove that the minimizer is unique

Equality can hold only if every nonnegative summand is zero. Then

$A_B y = b_B = A_B x^*$, and nonsingularity forces $y = x^*$. Therefore every other $y \in P$ satisfies $c^\top x^* < c^\top y$, so x^* is a vertex. $\square$

Only Finitely Many Corners (Cor. 2.1)

Corollary

A polyhedron defined by m linear constraints in $\mathbb{R}^n$ has finitely many basic solutions—at most $\binom{m}{n}$.

Proof idea. A basic solution is pinned by a choice of n linearly independent active constraints. Each choice pins at most one point, and there are $\binom{m}{n}$ choices. $\square$

Finite does not mean small

The unit cube in $\mathbb{R}^n$, described by $0 \leq x_i \leq 1$, has 2^n BFS.

n	cube corners 2^n
2	4
10	1,024
100	$\approx 1.3 \times 10^{30}$
333	> a googol

“Check every corner” is not an algorithm. Chapter 3 must be smarter.

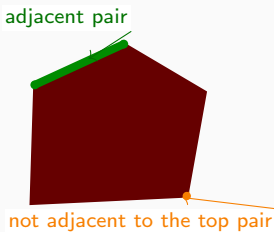
Adjacent Basic Solutions & Edges

Definition

Two distinct basic solutions in $\mathbb{R}^n$ are **adjacent** if they share $n - 1$ linearly independent active constraints.

If both are feasible, the segment joining them is an **edge** of P .

- In $\mathbb{R}^2$, adjacent corners share one active line—the edge between them.
- Adjacency means “one constraint swap away.” The simplex method walks edge by edge, as in the Lecture 2 grapher.



Summary & What's Next

- A polyhedron is a finite intersection of halfspaces, hence convex; a bounded polyhedron is a polytope.
- A corner has three equivalent descriptions: extreme point (geometry) = vertex (optimization) = BFS (algebra).
- Corners are pinned by n independent active constraints, so there are finitely many—possibly astronomically many.
- Adjacent corners share $n - 1$ active constraints, creating the edges that a future algorithm will walk.

Next lecture

Corners in standard form (bases), degeneracy, when corners exist at all, and the theorem that lets us search only corners.